

WASECA, MN, USA

WMO: 722032

Lat: 44.074N

Lon: 93.553W

Elev: 1126

StdP: 14.11

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 54916

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-7.5	-18.0	2.1	-10.9	-13.3	2.8	-7.2	32.0	21.6	28.6	18.3	5.5	240	0.591

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.8	88.0	72.7	84.3	70.8	81.9	69.1	75.9	83.5	74.2	81.5	72.4	79.5	12.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.1	128.0	81.0	72.1	123.8	79.9	70.2	116.0	77.7	40.1	84.5	38.5	82.3	37.0	79.6	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 26.6	23.9	20.6	DB	-19.2	94.5	5.5	3.2	-23.2	96.8	-26.4	98.7	-29.5	100.5	-33.4	102.9	(4)
(5)			WB	-19.7	80.0	5.4	2.8	-23.6	82.0	-26.7	83.7	-29.7	85.3	-33.6	87.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	45.4	16.4	18.3	32.6	46.1	57.7	67.4	70.4	67.6	62.2	48.0	35.1	21.3
	DBStd	21.63	12.96	12.13	12.25	10.89	9.21	6.07	5.57	5.41	8.05	9.32	10.80	11.45
	HDD50	4209	1042	887	552	196	30	0	0	0	6	150	458	888
	HDD65	7669	1507	1307	1004	568	265	43	14	33	144	533	898	1353
	CDD50	2535	0	0	14	80	268	521	633	547	373	89	10	0
	CDD65	522	0	0	1	2	38	114	183	115	62	7	0	0
	CDH74	4709	0	0	4	50	424	995	1684	926	543	81	2	0
	CDH80	1106	0	0	0	6	135	241	425	164	120	15	0	0
Wind	WSAvg	9.4	10.7	10.3	10.8	11.6	10.8	8.7	6.6	5.8	7.6	9.0	10.4	10.1
Precipitation	PrecAvg	34.8	1.0	0.9	2.2	3.1	4.1	5.2	4.5	4.8	3.4	2.6	1.9	1.3
	PrecMax	44.2	2.5	2.2	4.0	6.2	7.6	8.5	7.2	10.2	11.4	6.5	5.6	3.6
	PrecMin	23.0	0.2	0.1	0.3	0.8	1.6	3.1	1.7	1.7	1.0	0.6	0.2	0.2
	PrecStd	5.4	0.6	0.5	0.9	1.4	1.7	1.7	1.8	1.9	2.4	1.7	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.0	52.1	72.0	80.7	91.0	91.0	91.2	88.2	88.0	81.7	68.3	51.8
		MCWB	37.5	48.1	59.7	59.2	67.0	72.6	78.1	74.7	70.7	63.0	56.3	46.7
	2%	DB	38.6	41.3	62.7	73.0	82.3	86.0	87.9	83.8	82.4	73.5	60.8	41.5
		MCWB	35.6	38.0	53.8	57.8	63.0	69.2	75.3	74.2	68.8	62.2	50.5	37.9
	5%	DB	35.9	36.7	55.0	68.5	78.7	81.9	84.1	81.4	80.6	70.0	54.7	37.4
		MCWB	34.0	34.9	49.7	54.6	61.7	66.0	73.6	71.2	67.4	58.6	47.9	35.5
	10%	DB	33.9	34.3	49.8	63.2	72.9	79.3	81.7	79.2	77.1	63.3	51.6	36.0
		MCWB	32.3	32.8	45.7	51.6	58.9	65.3	71.3	69.7	65.9	53.6	45.4	34.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	47.9	62.2	62.4	70.8	75.5	80.7	77.3	74.6	70.6	57.3	45.6
		MCDB	41.7	51.9	70.9	72.6	83.5	85.8	89.6	84.6	82.8	77.1	63.8	49.9
	2%	WB	35.7	38.5	55.3	59.5	67.3	72.8	77.3	75.1	72.2	62.9	52.9	38.7
		MCDB	38.1	41.3	60.1	71.1	78.6	80.9	85.1	81.5	79.9	71.9	58.7	41.6
	5%	WB	33.8	35.0	49.7	56.5	64.4	70.9	74.9	73.4	70.5	58.8	48.3	36.1
		MCDB	35.3	36.6	55.8	65.9	74.7	78.8	82.4	80.0	77.2	66.6	55.0	37.8
	10%	WB	32.0	32.7	44.3	53.3	61.5	68.5	73.1	71.7	68.0	54.9	44.8	34.0
		MCDB	33.0	33.9	49.1	60.9	70.8	76.4	80.2	77.8	75.1	63.0	49.9	35.4
Mean Daily Temperature Range	5% DB	MDBR	15.2	16.5	15.3	19.3	20.4	19.5	19.8	19.6	21.1	19.8	16.2	13.8
		MCDBR	15.3	16.4	23.6	28.1	27.0	23.1	20.6	20.6	24.1	27.7	24.4	15.0
		MCWBR	13.8	13.9	16.2	16.3	12.7	11.1	11.6	12.3	12.6	15.9	15.8	13.0
	5% WB	MCDBR	14.7	15.6	21.3	24.3	21.4	19.2	19.0	18.0	20.0	23.0	22.7	14.1
		MCWBR	13.5	13.5	15.4	15.6	11.9	12.1	11.9	11.8	12.2	15.2	15.5	12.4
Clear-Sky Solar Irradiance	taub		0.267	0.287	0.313	0.368	0.394	0.410	0.412	0.404	0.374	0.323	0.289	0.267
	taud		2.435	2.370	2.422	2.314	2.304	2.300	2.292	2.305	2.359	2.499	2.513	2.454
	Ebn at Noon		277	288	294	284	279	273	271	269	269	272	266	266
	Edh at Noon		23	29	32	39	40	41	41	39	34	26	21	20
All-Sky Solar Radiation	RadAvg		561	902	1179	1459	1709	1911	2022	1693	1342	871	566	411
	RadStd		65	94	118	155	135	154	86	121	97	115	52	34

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (59 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page